

PYTHON PROGRAMMING ASSIGNMENT

Topic: Basic Data Structures, NumPy Arrays, Pseudocode & Numerical Simulations

Student Name: _____ Date: _____

Course / Class: _____

Assignment Instructions:

- Complete all questions by writing clean, well-commented Python code.
- Submit your script as a single .py file or a Jupyter Notebook (.ipynb).
- Ensure that all outputs match the requirements described in each question statement.

Question 1: List Filtering & Data Processing

SCENARIO

A local library wants to analyze the character lengths of book titles to optimize their digital display screens and catalog layout.

TASK REQUIREMENTS

Write a Python program that completes the following steps:

1. Define a list of book titles containing: ["Dune", "The Hobbit", "1984", "Foundation", "Hamlet"].
2. Iterate through the list using a loop and create a **new list** containing only the titles that have **more than 5 characters**.
3. Print the original list, the filtered list, and the total count of filtered titles with clear explanatory labels.

Question 2: Array Operations & Vectorization with NumPy

SCENARIO

A weather monitoring station records hourly temperature readings in Celsius for a given day:

```
celsius_readings = [18.5, 19.0, 21.2, 24.5, 26.8, 27.1, 25.4, 22.0]
```

TASK REQUIREMENTS

Using Python and the numpy library:

1. Convert the standard Python list of Celsius readings into a NumPy array.
2. Convert all temperatures to **Fahrenheit** using vectorized array operations and the formula:

$$F = (C \times 1.8) + 32$$

3. Perform and print calculations for the following statistics:
 - The average daily temperature in Fahrenheit.
 - The maximum (highest) temperature recorded in Fahrenheit.
 - A filtered array containing only the readings strictly above **75°F** using boolean masking.

Question 3: Pseudocode Conversion & Function Logic

SCENARIO

An online store applies discounts based on customer membership tiers and order totals. The store manager provided the pseudocode below specifying the automated checkout logic.

PROVIDED PSEUDOCODE

```
BEGIN
  INPUT customer_tier (string: "Gold", "Silver", or "Standard")
  INPUT total_amount (float)

  IF customer_tier IS EQUAL TO "Gold" THEN
    SET discount_rate = 0.20
  ELSE IF customer_tier IS EQUAL TO "Silver" THEN
    SET discount_rate = 0.10
  ELSE
    SET discount_rate = 0.05
  ENDIF

  IF total_amount IS GREATER THAN 100.0 THEN
    SET discount_rate = discount_rate + 0.05
  ENDIF

  CALCULATE final_price = total_amount * (1.0 - discount_rate)
  OUTPUT "Final total: " concatenated with final_price
END
```

TASK REQUIREMENTS

1. Translate the above pseudocode into a Python function named `calculate_checkout(customer_tier, total_amount)`.
2. Test your function with the following sample inputs:
 - Tier: "Gold", Total: \$120.0
 - Tier: "Standard", Total: \$50.0
3. Return and print the final calculated total formatted to 2 decimal places (e.g., \$90.00).

Question 4: Random Growth Sequence Simulation

CONCEPT BACKGROUND

Consider a modified sequence where each new term is based on the previous term, but with a randomly generated multiplier between 0.5 and 1.5 at each step.

TASK REQUIREMENTS

1. Start with an initial value of 100 and generate a sequence of 30 steps.
2. At each step, calculate the next value as:

$$\text{next_value} = \text{current_value} \times \text{random_factor}$$

where `random_factor` is a random floating-point number in the range `[0.5, 1.5]`.

3. Store the sequence in a NumPy array or list and plot the values over time (iterations) using Matplotlib.

Question 5: Gauss-Seidel Method for Linear Equations

CONCEPT & PSEUDOCODE

The **Gauss-Seidel method** is an iterative technique used to solve a system of linear equations $\mathbf{Ax} = \mathbf{b}$. Given a system of 3 equations with 3 unknowns:

$$a_{11}x + a_{12}y + a_{13}z = b_1$$

$$a_{21}x + a_{22}y + a_{23}z = b_2$$

$$a_{31}x + a_{32}y + a_{33}z = b_3$$

PROVIDED PSEUDOCODE

```
BEGIN
  INITIALIZE x = 0.0, y = 0.0, z = 0.0
  SET tolerance = 0.0001
  SET max_iterations = 50

  FOR iteration FROM 1 TO max_iterations DO
    SET x_old = x
    SET y_old = y
    SET z_old = z

    // Update x, y, z using the latest available values
    COMPUTE x = (b1 - a12 * y - a13 * z) / a11
    COMPUTE y = (b2 - a21 * x - a23 * z) / a22
    COMPUTE z = (b3 - a31 * x - a32 * y) / a33

    // Check for convergence
    IF |x - x_old| < tolerance AND |y - y_old| < tolerance AND |z - z_old| <
tolerance THEN
      OUTPUT "Converged in iteration: " + iteration
      BREAK
    ENDIF
  ENDFOR

  OUTPUT x, y, z
END
```

SAMPLE SYSTEM OF EQUATIONS TO SOLVE

$$\begin{aligned} 10x + 1y - 1z &= 11 \\ 1x + 10y + 1z &= 12 \\ 2x - 1y + 10z &= 21 \end{aligned}$$

TASK REQUIREMENTS

1. Translate the Gauss-Seidel pseudocode into a Python program.
2. Solve the provided 3×3 system of equations using initial guesses $x_0 = 0, y_0 = 0, z_0 = 0$.
3. Print the final values of x, y , and z rounded to 4 decimal places along with the total number of iterations required to reach convergence.